

Thoshith R R 250701797-2025-CSE ▾T2**Started on** Saturday, 3 October 2026, 11:00 AM**State** Finished**Completed on** Saturday, 3 October 2026, 11:06 AM**Time taken** 5 mins 20 secs**Grade** 10.00 out of 10.00 (100%)

**Question 1** | Correct Mark 10.00 out of 10.00**Playing with Numbers:**

Ram and Sita are playing with numbers by giving puzzles to each other. Now it was Ram term, so he gave Sita a positive integer 'n' and two numbers 1 and 3. He asked her to find the possible ways by which the number n can be represented using 1 and 3. Write any efficient algorithm to find the possible ways.

**Example 1:**

**Input:** 6

**Output:** 6

**Explanation:** There are 6 ways to 6 represent number with 1 and 3

1+1+1+1+1+1

3+3

1+1+1+3

1+1+3+1

1+3+1+1

3+1+1+1

**Input Format**

First Line contains the number n

**Output Format**

**Print:** The number of possible ways 'n' can be represented using 1 and 3

Sample Input

6

Sample Output

6

**Answer:** (penalty regime: 0 %)

```
1 #include <stdio.h>
2
3 int main() {
4     int n;
5     scanf("%d", &n);
6
7     long long dp[n + 1];
8
9     dp[0] = 1;
10
11     for (int i = 1; i <= n; i++) {
12         dp[i] = dp[i - 1];
13
14         if (i >= 3) {
15             dp[i] += dp[i - 3];
16         }
17     }
18
19     printf("%lld", dp[n]);
20
21     return 0;
22 }
```

|   | Input | Expected          | Got               |   |
|---|-------|-------------------|-------------------|---|
| ✓ | 6     | 6                 | 6                 | ✓ |
| ✓ | 25    | 8641              | 8641              | ✓ |
| ✓ | 100   | 24382819596721629 | 24382819596721629 | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 10.00/10.00.

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